

Week 3 Practical (QGIS) – Submission Package

Data used: municipalities.shp, BC250_Terra_Indigena_A.shp, MCMV_new.shp (from the provided ZIP folders).

Task	Result (for report/marking)
1. São Paulo centroids + boundary	Created centroid layer for SP municipalities and dissolved SP boundary. See Map 1.
2. Mean HDI by state	Computed mean of IDHM_10 by UF. See attached CSV: Q2_mean_HDI_by_state.csv
3. Gaúcha do Norte $\cap$ Parque do Xingu	Intersection polygon created. See Map 2.
4. Acre closest projects + 10km buffers	Closest pair in AC: Project_ID 18 and 20 ($\approx$ 23.7 km apart). 10 km buffer made around
5. Lowest & highest total UH by municipality	Highest total UH: Rio de Janeiro (RJ) = 46,966. Lowest non-zero: Presidente Figue

Map 1 – São Paulo centroids and state boundary

São Paulo: Municipal Centroids and State Boundary



Boundary is the dissolved São Paulo municipality polygons; points are municipal centroids.

Map 2 – Gaúcha do Norte area inside Parque do Xingu

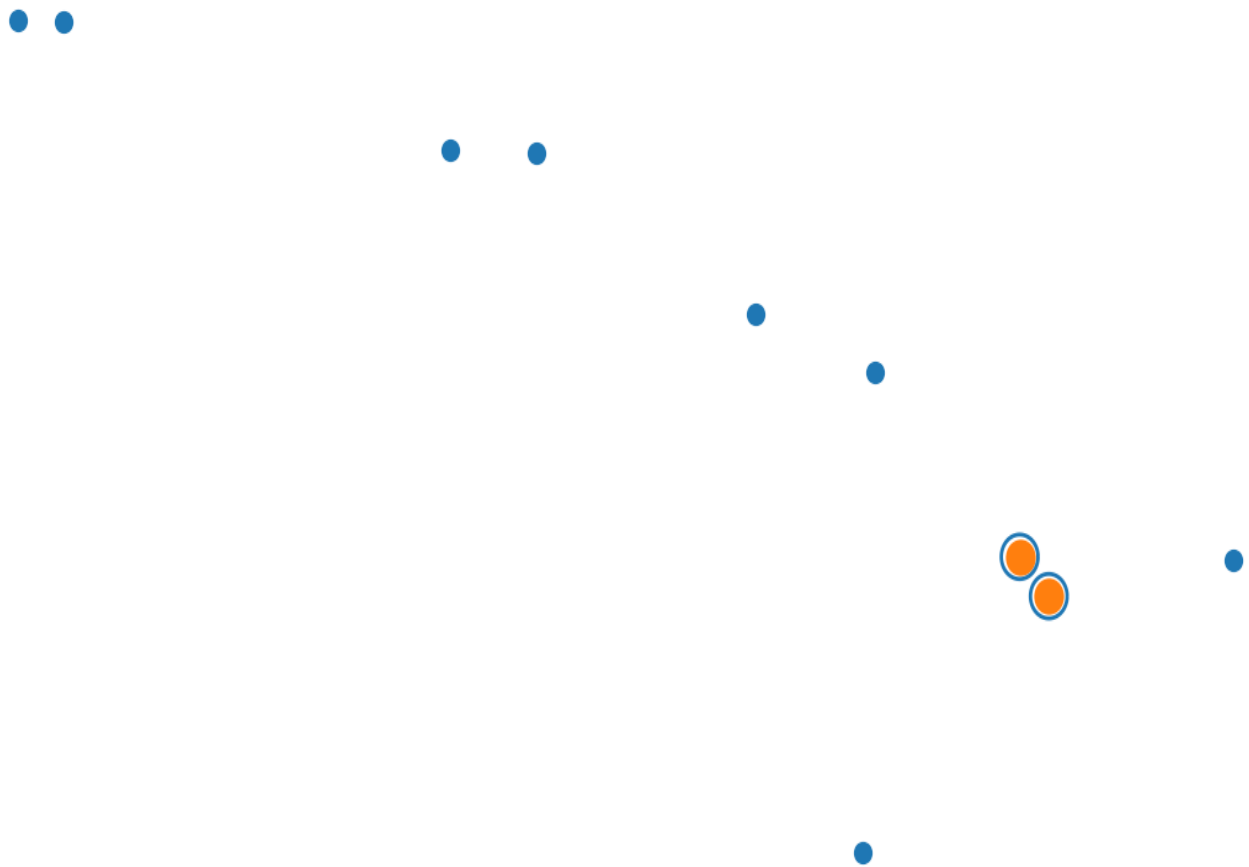
Intersection: Gaúcha do Norte within Parque do Xingu



Dashed outline = Gaúcha do Norte municipality; solid outline = Parque do Xingu; filled = intersection.

Map 3 – Acre closest projects and 10 km buffers

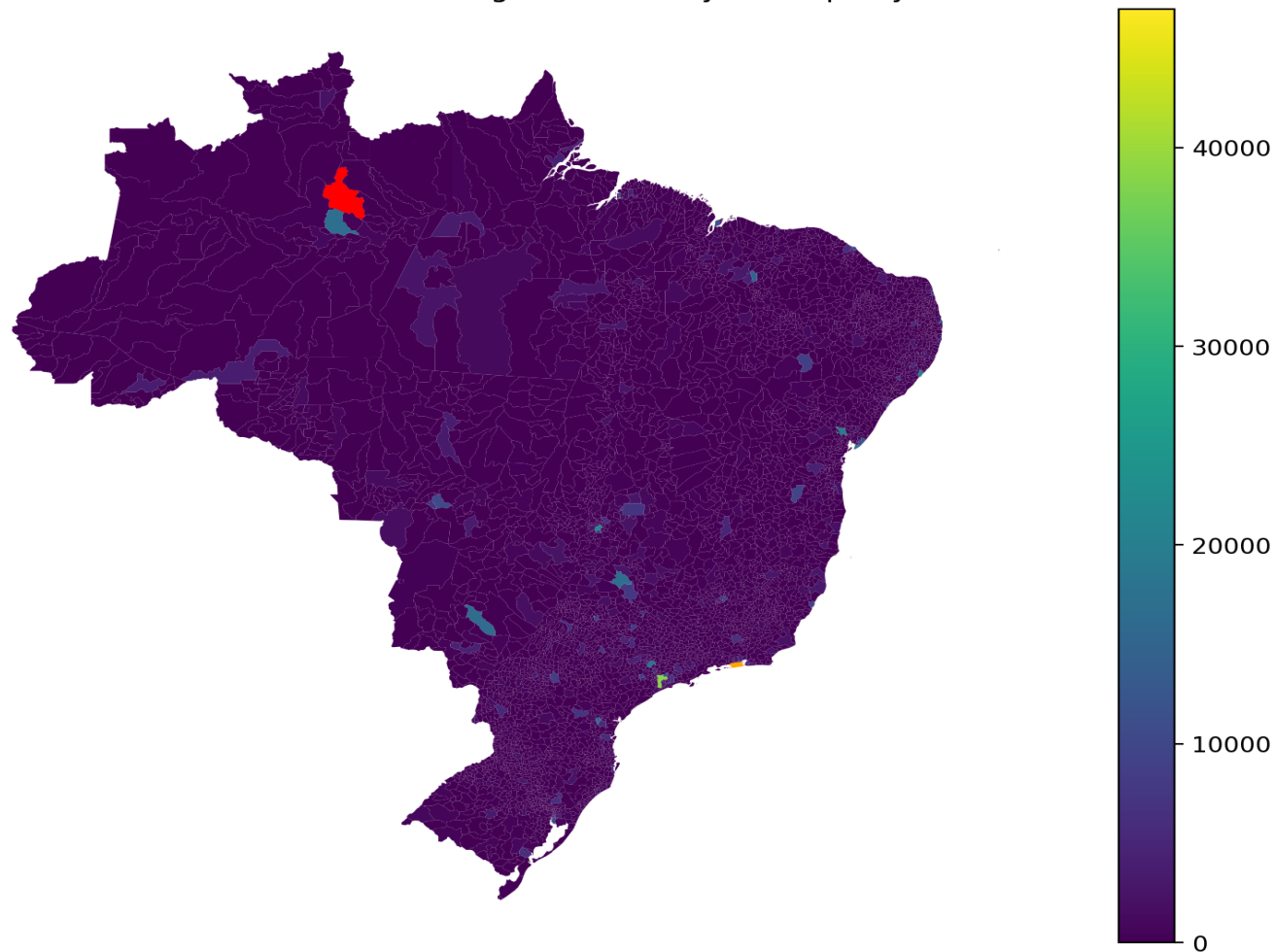
Acre (AC): Two Closest Projects and 10 km Buffers



Two closest projects (Project_ID 18 and 20) with 10 km buffers (distance calculations in EPSG:31979).

Map 4 – Total MCMV UH by municipality (choropleth)

Brazil: Total MCMV Housing Units (UH) by Municipality



Choropleth shows sum of UH per municipality. Red mark indicates lowest non-zero (Presidente Figueiredo, AM). Orange mark indicates highest (Rio de Janeiro, RJ).